

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)**REVIEW TEST****06-09-2020****JEE(Main + Advanced) : NURTURE COURSE [PHASE : PHASE-TNPS, TOPS & TRPS]****PAPER-2****Time : 3 Hours****Maximum Marks : 180****READ THE INSTRUCTIONS CAREFULLY****GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR**Please see the last page of this booklet for rest of the instructions**

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

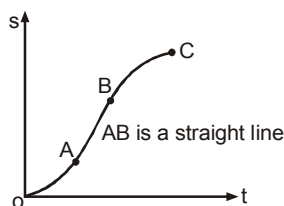
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

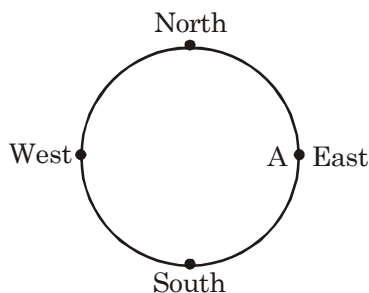
1. Displacement time graph of a particle moving in a straight line is as shown in figure. Select the correct alternative (s) :



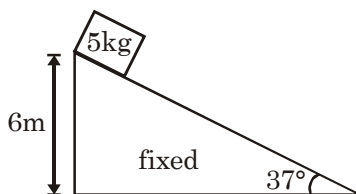
- (A) Work done by all the force in region OA and BC is positive.
- (B) Work done by all the force in region AB is zero.
- (C) Work done by all the force in region BC is negative.
- (D) Work done by all the force in region OA is negative.

Space for Rough Work

2. A car travels on a flat circular path of radius 250 m with constant speed 20 m/s :-



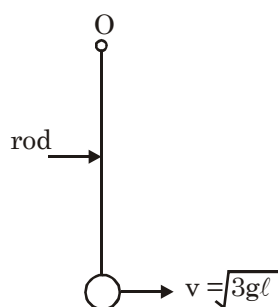
- (A) When car is at A, its acceleration is 1.6 m/s^2 in the west direction.
 (B) Acceleration of car at A will depend on the sense in which car is moving i.e. clockwise or anticlockwise.
 (C) Frictional force acting on car will always be towards centre of circle.
 (D) Velocity of car remains constant as it moves on circle.
3. A block is released from rest on a fixed incline plane from a height 6m above the ground as shown in figure where coefficient of friction between block and incline plane is 0.5. Choose the correct option(s) for motion of block from shown position to the bottom of incline :-



- (A) Change in potential energy is -300 J .
 (B) Work done by normal is 400 J .
 (C) Work done by friction is -200 J .
 (D) Final velocity of block when it just reaches to bottom of incline plane is $2\sqrt{10} \text{ m/s}$.

Space for Rough Work

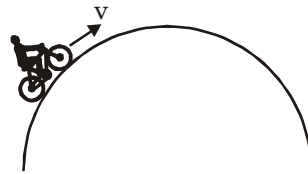
4. Potential energy of a system is given by $U(x) = (x+1)(x+2)$. Then
- (A) point $x = -\frac{3}{2}$ corresponds to equilibrium position of the system.
- (B) point $x = -1$ and $x = -2$ corresponds to equilibrium position of the system.
- (C) system is in stable equilibrium position at the $x = -\frac{3}{2}$
- (D) system is in unstable equilibrium position at $x = -\frac{3}{2}$
5. A particle of mass 'm' is connected with a massless rod of length ' ℓ '. Velocity of $\sqrt{3g\ell}$ is given to the particle when it is at lower most point. Particle moves around point O in circular path. Select the **CORRECT** statement(s) :-



- (A) Maximum height reached by the particle is $\frac{4\ell}{3}$ from initial lowest position.
- (B) Maximum height reached by the particle is $\frac{3\ell}{2}$ from initial lowest position.
- (C) Acceleration of particle at heighest point is $\frac{g}{2} \text{ m/s}^2$.
- (D) Acceleration of particle at heighest point is $\frac{\sqrt{3}g}{2} \text{ m/s}^2$.

Space for Rough Work

6. On a circular hill, a cyclist is going up with constant speed as shown in diagram. Choose the **CORRECT** statement(s) :



- (A) Net friction force will act tangentially downward at shown instant.
- (B) Net friction force will act tangentially upward at shown instant.
- (C) Net friction force will be zero at shown instant.
- (D) As it goes up, normal reaction increases

Space for Rough Work

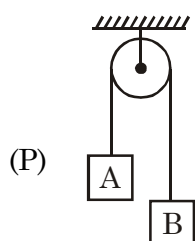
SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

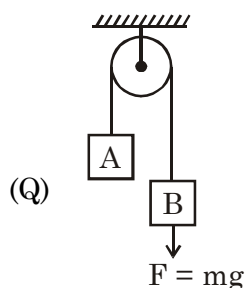
7. In the list-I there are few pulley block systems, under different conditions and list-II has the magnitude of acceleration of blocks. (Given mass of A = m, mass of B = 2m)

List-I

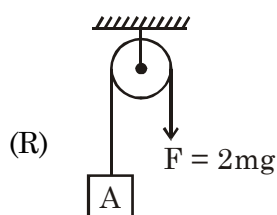
List-II



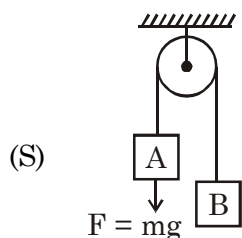
(1) 0



(2) $\frac{g}{4}$



(3) $\frac{g}{3}$



(4) g

(5) $\frac{2g}{3}$

(A) P → 1, Q → 4, R → 3, S → 2

(B) P → 3, Q → 2, R → 1, S → 4

(C) P → 3, Q → 5, R → 4, S → 1

(D) P → 5, Q → 4, R → 3, S → 1

Space for Rough Work

8. A ball is thrown from point O with speed 50 m/s from ground at an angle 53° with horizontal it lands on same horizontal level at point A and a dog is running with 20 m/s constant velocity along OA.

List-I

List-II

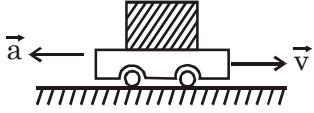
(values are in appropriate SI unit)

- | | |
|--|--|
| (P) Range of ball on ground | (1) 240 |
| (Q) Horizontal displacement of ball w.r.t. dog after 8 sec. | (2) 80 |
| (R) Magnitude of velocity of ball w.r.t. dog after 2sec. | (3) $10\sqrt{5}$ |
| (S) Magnitude of velocity of ball w.r.t. dog after 5 sec. | (4) $10\sqrt{2}$ |
| (A) $P \rightarrow 2, Q \rightarrow 4, R \rightarrow 1, S \rightarrow 3$ | (B) $P \rightarrow 1, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 4$ |
| (C) $P \rightarrow 3, Q \rightarrow 2, R \rightarrow 1, S \rightarrow 4$ | (D) $P \rightarrow 4, Q \rightarrow 1, R \rightarrow 2, S \rightarrow 3$ |

9. List-I contains different situations of trolley moving on a straight horizontal road and list-II contains the direction of total contact force on the block (Assume that there is no slipping between block and trolley)

List-I

List-II

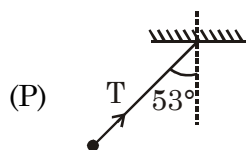
- | | |
|---|---|
| (P)  Trolley at rest | (1)  |
| (Q)  Trolley moving with constant velocity | (2)  |
| (R)  | (3)  |
| (S)  | (4)  |
| | (5)  |
| (A) $P \rightarrow 2, Q \rightarrow 4, R \rightarrow 1, S \rightarrow 3$ | (B) $P \rightarrow 2, Q \rightarrow 2, R \rightarrow 1, S \rightarrow 3$ |
| (C) $P \rightarrow 1, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 4$ | (D) $P \rightarrow 1, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 4$ |

Space for Rough Work

10. List-I shows different cases of a particle and list-II contains values of tension in string at the instants shown. (Assume mass of particle is equal to m)

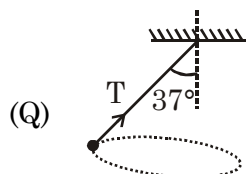
List-I

List-II



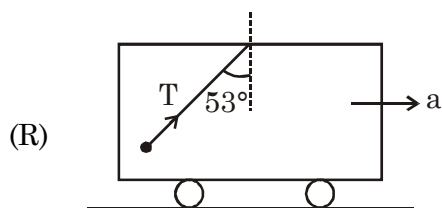
(1) $\frac{4mg}{5}$

Extreme position in simple pendulum



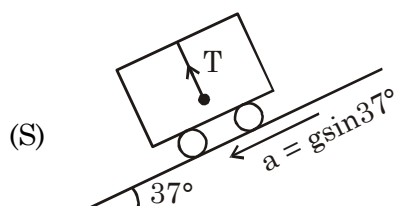
(2) $\frac{5mg}{4}$

Conical pendulum



(3) $\frac{3mg}{5}$

Cart is accelerated such that particle remains at rest as shown w.r.t. cart



(4) $\frac{5mg}{3}$

Cart is accelerating freely on the incline.

Particle remains at rest w.r.t. cart as shown.

(A) $P \rightarrow 3, Q \rightarrow 2, R \rightarrow 4, S \rightarrow 1$

(B) $P \rightarrow 2, Q \rightarrow 4, R \rightarrow 3, S \rightarrow 1$

(C) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 1, S \rightarrow 2$

(D) $P \rightarrow 3, Q \rightarrow 4, R \rightarrow 1, S \rightarrow 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

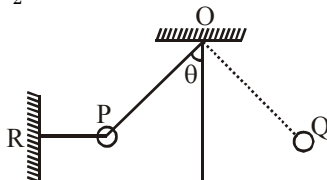
- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

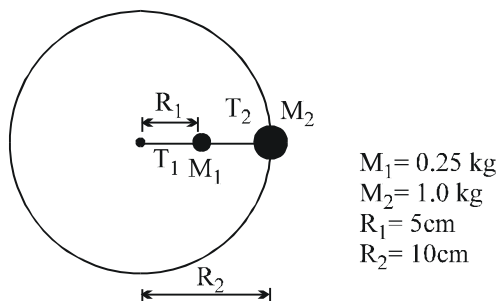
-									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A ball of mass 1kg is at rest in position P by means of two light strings OP and RP as shown in the figure. The string RP is now cut and the ball swings upto a position Q. If $\theta = 60^\circ$ and tension in the string connected to point O in position OP and OQ are T_1 (just before cutting) and T_2 respectively, then $(T_1 + T_2)$ is equal to :-

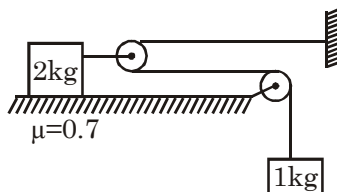


2. Two different masses are connected to two **light and inextensible** strings as shown in the figure. Both masses rotate about a central fixed point with constant angular speed of 10 rad s^{-1} on a smooth horizontal plane. Find the ratio of tensions $\frac{T_1}{T_2}$ in the strings.

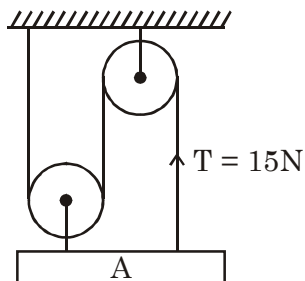


Space for Rough Work

3. Find the velocity (in m/s) of 2kg mass when 1kg block has moved 1m down. The entire system starts from rest with string taut.



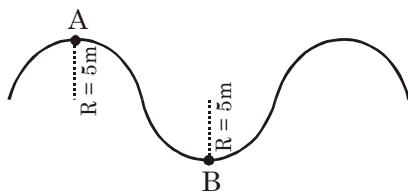
4. In the arrangement shown, block A is in equilibrium. Find mass of block A (in kg) ($g = 10 \text{ m/s}^2$)



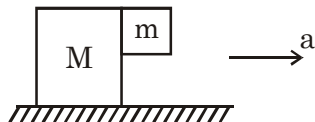
5. The potential energy of a body is given by $U = (40 + x^2 + y^2 + z^2) \text{ J}$, where x, y, z in meter. Calculate the magnitude of the conservative force associated with the potential energy when the body is at position (1.0 m, 0.5 m, -1.0 m).

Space for Rough Work

6. A toy car of mass 1 kg moving with constant speed 5 m/s on a frictionless road as shown in figure. Normal reaction by road on car N_A & N_B at position A and B. If $N_B - N_A = \alpha$ (in N) the value of α will be.



7. With what minimum acceleration (in m/s^2) mass M must be moved on frictionless surface so that m remains at rest with respect to 'M' as shown. The coefficient of friction between M and m is 0.4. ($g = 10 \text{ m/s}^2$)



8. A large sphere of radius R (in m) fixed from ground as shown in figure. A particle of mass m released from top point of sphere. Height from the ground where particle m will leave contact with surface of sphere is xR . Find the value of x. (Assume all surfaces are frictionless.)



Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Select **CORRECT** order/s of boiling point.

- (A) $H_2 > He$ (B) $D_2O > H_2O$ (C) $SbH_3 < NH_3$ (D) $HBr < HCl$

2. Select **INCORRECT** order/s for given properties.

- (A) PO bond length $PO_4^{3-} < HPO_3^{2-} < H_2PO_2^-$

- (B) PO bond order $PO_4^{3-} > HPO_3^{2-} > H_2PO_2^-$

- (C) OPO bond angle $PO_4^{3-} < HPO_3^{2-} < H_2PO_2^-$

- (D) Ionisable hydrogen $PO_4^{3-} < HPO_3^{2-} < H_2PO_2^-$

Space for Rough Work

3. 20 ml of a mixture of C_3H_8 and CO when burnt completely in oxygen produces 40 ml CO_2 gas. The correct statements are -
- (A) 15 ml of CO is present in original mixture
 - (B) 10 ml of C_3H_8 is present in original mixture
 - (C) 10 ml of CO is present in original mixture
 - (D) 55 ml of O_2 is consumed for combustion
4. Sodium hypochlorite (NaOCl) solution is commonly used as sanitiser in covid-19 pandemic. To prepare 1 litre sanitiser, 5 ml of 5% w/v NaOCl solution is used. Which of following is/are correct statement(s) ?
- (A) 1 litre of sanitiser solution contains 5 g of NaOCl
 - (B) 1 litre of sanitiser solution contains 0.25 g of NaOCl
 - (C) Molarity of sanitiser w.r.t. NaOCl is 'M' then $M/0.01$ is equal to 0.33
 - (D) Molarity of 5% w/v solution of NaOCl is 0.67.

Space for Rough Work

5. For the reaction : $\text{PbS} + \text{H}_2\text{O}_2 \longrightarrow \text{PbSO}_4 + \text{H}_2\text{O}$;
which are correct about the equivalent weight of respective species ?
- (A) $E_{\text{PbS}} = \frac{M}{8}$ (B) $E_{\text{H}_2\text{O}_2} = M$
(C) $E_{\text{PbS}} = \frac{M}{2}$ (D) $E_{\text{H}_2\text{O}_2} = M / 2$
6. An oleum sample has SO_3 & H_2SO_4 in 1 : 3 mass ratio. Select the correct statement(s)
- (A) % labelling of sample is 105.625 %
(B) % labelling of sample is 109 %
(C) If 3g of H_2O is added to 100 g of above sample, new labelling would be 102.55%
(D) If 2g of H_2O is added to 100 g of above sample, new labelling would be 106%

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match the list:

List- I (Molecular species)	List- II (Properties)
(P) ClO_4^-	(1) Planar and polar
(Q) NH_2^-	(2) All bond lengths are identical
(R) SOF_4	(3) Axial d-orbitals participates in hybridisation
(S) XeOF_4	(4) Non planar and non polar
	(5) p_π - d_π bond(s) are present

The correct option is

- | | |
|--|--|
| (A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 5$ | (B) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 5; S \rightarrow 3$ |
| (C) $P \rightarrow 2; Q \rightarrow 5; R \rightarrow 2; S \rightarrow 1$ | (D) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$ |

Space for Rough Work

8. Given $x = \% \text{ increase in volume (At constant P \& T)}$

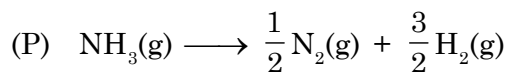
$M_1 = \text{Theoretical molar mass, } M_2 = \text{Observed molar mass of gases mixture}$

$D_1 = \text{Theoretical vapour density, } D_2 = \text{Observed vapour density of gases mixture}$

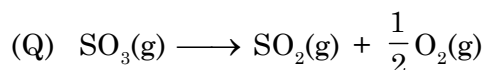
List-I

List-II

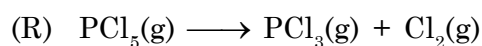
(Extent of reaction = 100%)



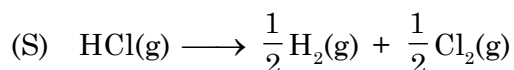
(1) $x = 33.33\%$



(2) $x = 25\%$



(3) $x = 100\%$



(4) $x = 50\%$

(5) $M_1 > M_2$

(6) $D_1 = D_2$

The correct option is

(A) $P \rightarrow 3,5; Q \rightarrow 4,5; R \rightarrow 3,5; S \rightarrow 6$

(B) $P \rightarrow 2,5; Q \rightarrow 4,5; R \rightarrow 4,5; S \rightarrow 6$

(C) $P \rightarrow 3,5; Q \rightarrow 6; R \rightarrow 4,5; S \rightarrow 2,5$

(D) $P \rightarrow 4,5; Q \rightarrow 4,5; R \rightarrow 2,5; S \rightarrow 6$

Space for Rough Work

9. List-I

- (P) Mass of one molecule of H_2SO_4
(Q) Molar mass of H_2SO_4
(R) 1 amu
(S) Relative atomic mass of Mg

The correct option is

- (A) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 4$; $S \rightarrow 1$
(B) $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 4$; $S \rightarrow 1$
(C) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 1$
(D) $P \rightarrow 1$; $Q \rightarrow 4$; $R \rightarrow 2$; $S \rightarrow 3$

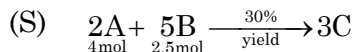
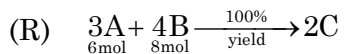
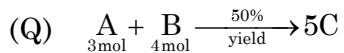
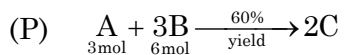
List-II

- (1) 24
(2) 98 amu
(3) 98 gm
(4) $1.66 \times 10^{-27}\text{kg}$

Space for Rough Work

10. List-I

(Reactions)



List-II

(1) No reactant is left

(2) A is in excess and 2.4 mole of C is produced

(3) A is in excess and 0.45 mole of C is produced

(4) B is in excess and 7.5 mole of C is produced

The **CORRECT** option is :

(A) $\text{P} \rightarrow 2$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 3$; $\text{S} \rightarrow 1$

(B) $\text{P} \rightarrow 3$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 2$; $\text{S} \rightarrow 4$

(C) $\text{P} \rightarrow 2$; $\text{Q} \rightarrow 4$; $\text{R} \rightarrow 1$; $\text{S} \rightarrow 3$

(D) $\text{P} \rightarrow 2$; $\text{Q} \rightarrow 1$; $\text{R} \rightarrow 4$; $\text{S} \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

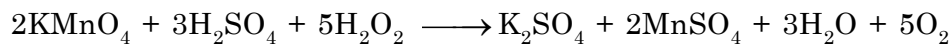
-									
●	●	●	0	0	0	●	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Find total number of $p_\pi-d_\pi$ bonds present in CO_3^{2-} , SO_3 , SO_4^{2-} and PO_4^{3-}
- In the reaction : $\text{Cl}_2 + \text{KOH} \rightarrow \text{KCl} + \text{KClO}_3 + \text{H}_2\text{O}$
The equivalent weight of H_2O is-

Space for Rough Work

3. A sample of 125 ml H_2O_2 (aq) solution required 200 ml 0.1M KMnO_4 for the complete reaction. Calculate volume strength of H_2O_2 at STP (1bar & 0°C .)



4. 2 moles of a hydrocarbon C_xH_y on complete combustion gives 8 moles CO_2 and 4 moles of H_2O then value of $\frac{x+y}{2}$ will be.

5. 1.6 g of an organic compound containing sulphur, when treated with series of reagents, produces H_2SO_4 which on reaction with BaCl_2 produces 0.233 g of BaSO_4 . Calculate % by mass of S in the organic compound.

(Given : Atomic weight of Ba = 137)

Space for Rough Work

6. 100 gm oleum sample (label : 107.8 %) is mixed with 7.8 gm water and the resulting product required 1.1 ℓ of y M NaOH solution for complete neutralization. The value of 'y' is.
7. 500 ml 33.6 V H_2O_2 solution is left open due to which some H_2O_2 decomposes and mass of O_2 gas evolved is 8gm. Find new volume strength of left H_2O_2 solution. [Volume of solution remains constant] (Volume strength consider at 1 atm, 273 K)
8. A solution of A (MM = 20) and B (MM = 10), $[X_B = 0.6]$ having density = 0.7 gm/mL, then the molarity of B in this solution.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. In a ΔABC , if $\sin A$, $\sin B$, $\sin C$ are in A.P. then
 (A) a , b , c are also in A.P. (B) a , b , c are in H.P.
 (C) $\tan \frac{A}{2} \tan \frac{C}{2} = \frac{1}{3}$ (D) $\cot \frac{A}{2}$, $\cot \frac{B}{2}$, $\cot \frac{C}{2}$ are also in A.P.
2. Let $(x - 2)^2 + (y - 3)^2 + (z + 1)^2 = 0$ then $\begin{vmatrix} x & 5 & y \\ z & 0 & 1 \\ -3 & 2 & z^2 \end{vmatrix}$ is
 (A) positive (B) negative (C) equal to 7 (D) equal to -20
-

Space for Rough Work

3. Let two points $A(1, x)$ & $B\left(3, -\frac{1}{x}\right)$ are given then least possible value of AB is
 (A) greater than 2 (B) less than 5
 (C) a rational number (D) can't be determined
4. Let $x \in [0, 3\pi]$ and the equation is $\sin x \cdot \cos 2x = \cos x \cdot \cos 2x$ then
 (A) number of distinct solutions is 6 (B) number of distinct solutions is 9
 (C) sum of all distinct solutions is 9π (D) sum of all distinct solutions is 18π
5. Let $x^3 - x^2 + 3x - 2 = 0$ is having the roots α, β, γ then
 (A) $\alpha^2 + \beta^2 + \gamma^2 = -5$ (B) $(\alpha + \beta + \gamma) - (\alpha\beta\gamma) = -2$
 (C) $\alpha^2 + \beta^2 + \gamma^2 = 7$ (D) $(\alpha + \beta + \gamma) - \alpha\beta\gamma = 10$
6. $\sum_{r=2}^{\infty} \frac{(r^4 + 3r^2 + 10r + 10)}{2^r(r^4 + 4)}$ is
 (A) 1 (B) $\frac{11}{10}$ (C) $\frac{3}{2}$ (D) 4

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Match List-I with List-II and select the correct answer using the code given below the list.

List-I	List-II
(P) $\begin{vmatrix} 0 & (b-a) & (c-a) \\ (a-b) & 0 & (c-b) \\ (a-c) & (b-c) & 0 \end{vmatrix}$ less than	(1) 1
(Q) If $3\sin\theta + 5\cos\theta = 5$ then $ 5\sin\theta - 3\cos\theta $ is greater than or equals to	(2) 2
(R) If $x \in [0, 2\pi]$ then number of solutions of the equations $\sin^2 x - \cos x = \frac{1}{4}$ is	(3) 3
(S) In $\triangle ABC$ (with usual notations), $\frac{(r_1 + r_2)}{(1 + \cos C)} \times \frac{2\Delta}{abc}$ is divisor of	(4) 4
	(5) 5
(A) $P \rightarrow 1,2,3,4,5$; $Q \rightarrow 1,2,3$; $R \rightarrow 2$; $S \rightarrow 1$	(B) $P \rightarrow 1,2,3,4,5$; $Q \rightarrow 1,2,3$; $R \rightarrow 2$; $S \rightarrow 1,2,3,4,5$
(C) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 4,5$	(D) $P \rightarrow 1,2$; $Q \rightarrow 1,2$; $R \rightarrow 3,4$; $S \rightarrow 5$

Space for Rough Work

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

- | | |
|---|--------|
| (P) Sum of all possible integral solutions of the inequality $(x-1)^2 - 7 x-1 + 10 \leq 0$ is | (1) 0 |
| (Q) Let $1 + 2\sin x + 3\sin^2 x + \dots$ + upto infinite terms is equal to 4, then number of values of x in $\left[-\frac{3\pi}{2}, 4\pi\right]$ is | (2) 3 |
| (R) In a triangle if the length of two longer sides are 8 and 7 and its angles are in A.P., then smaller side can be | (3) 5 |
| (S) If the value of $1 + (2+2) + (3+3+3) + \dots + \left(\underbrace{11+11+\dots+11}_{11 \text{ times}}\right)$ is λ then $\frac{\lambda}{22}$ is | (4) 8 |
| | (5) 23 |

(A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 2,3; S \rightarrow 5$

(B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 5; S \rightarrow 2,3$

(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2,4; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

Space for Rough Work

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

- (P) $\left(\frac{3 \tan 20^\circ - \tan^3 20^\circ}{1 - 3 \tan^2 20^\circ} \right)^2$ is (1) 2
- (Q) Let $f(x) = ax^2 - bx - c$, ($a, b, c \in \mathbb{N}$) has minimum value of -4 at $x = 1$. If a, b, c are in A.P., then abc is not equal to (2) 3
- (R) If the sum of all solution(s) of the equation (3) 6
- $$\cos^2 \left(x + \frac{\pi}{6} \right) + \cos^2 \frac{\pi}{6} = \sin^2 \frac{\pi}{3} + \cos \left(x + \frac{\pi}{6} \right)$$
- (4) 9
- where $x \in (-\pi, \pi)$, is $k\pi$ and number of solution(s) of given equation is n , then $|2nk|$ is (5) 10
- (S) In ΔABC , let G is centroid and area of ΔAGB is 3 then area of ΔABC is (6) 11
- (A) $P \rightarrow 2; Q \rightarrow 1,2,4,5,6; R \rightarrow 3; S \rightarrow 1$ (B) $P \rightarrow 1,2; Q \rightarrow 1,3,4,5,6; R \rightarrow 3; S \rightarrow 1$
- (C) $P \rightarrow 1; Q \rightarrow 1,3,4,6; R \rightarrow 4; S \rightarrow 3$ (D) $P \rightarrow 2; Q \rightarrow 1,2,4,5,6; R \rightarrow 2; S \rightarrow 4$

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I

List-II

(P) Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_i, B_i, C_i are co-factor of (1) 0

a_i, b_i, c_i respectively and $\Delta' = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ then $\frac{\Delta^2}{\Delta'}$ is (2) 1

(Q) If $x > 0$ such that $(\sqrt{5})^{\log_5 x} (\sqrt{5})^{\log_{5^2} x} (\sqrt{5})^{\log_{5^4} x} (\sqrt{5})^{\log_{5^8} x} \dots \infty = 7$ (3) 3
then $\frac{x}{3.5}$ is

(R) Number of solution(s) of the equation $\sin x + \cos x - 1 = \sin 2x - 2$ is $[0, 3\pi]$ is (4) 2

(S) Minimum value of $\frac{(x^2 + y^2)(y^2 + z^2)(z^2 + x^2)}{8x^2 y^2 z^2}$ is

(A) $P \rightarrow 1; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

(D) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Let L_1 is maximum value of $3\sin x + 4\cos x$, L_2 is minimum value of $25\tan^2\theta + 16\cot^2\theta$ and L_3 is maximum value of $3\sin^2x + 4\cos^2x$ then $\frac{(L_1 + L_2 + L_3)}{10}$ is
- If $3\sqrt{3}$, $2\sqrt{2}\sin^2\theta$, $\sin\theta$, ($\theta \neq n\pi$, $n \in \mathbb{I}$) are in geometric progression, then number of values of θ in $[-2\pi, 3\pi]$ is
- In $\triangle ABC$, $\frac{\sin A}{2} = \frac{\sin B}{2} = \frac{\sin C}{1}$, then $2(\cos A + \cos B + \cos C)$ is
- Let a , b , c , d be four distinct positive real numbers that satisfy the equations.

$$(a^{2016} - c^{2016})(a^{2016} - d^{2016}) = 2017 \text{ and } (b^{2016} - c^{2016})(b^{2016} - d^{2016}) = 2017 \text{ then } \left| \frac{(cd)^{2016} - (ab)^{2016}}{4034} \right| \text{ is}$$

Space for Rough Work

5. Let $x + ky + z = 0$, $2x + 3y - 2kz = 0$ and $x - y - z = 0$ has a non-trivial solution the number of possible real value(s) of k is
6. In a ΔABC (with usual notations) let $a : b : c = 4 : 5 : 6$ then $\frac{8r}{7R}$ is
7. Let ΔABC is not a right angled triangle then $\frac{1}{8} \begin{vmatrix} \tan A & 1 & 1 \\ 1 & \tan B & 1 \\ 1 & 1 & \tan C \end{vmatrix}$ is
8. Let λ_1 is sum of values of x & λ_2 is product of values of x satisfying the equation $\begin{vmatrix} 1 & 1 & x \\ p+1 & p+1 & p+x \\ 3 & x+1 & x+2 \end{vmatrix} = 0$ then $\frac{\lambda_1}{\lambda_2}$ is (where $p \neq 0$)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator